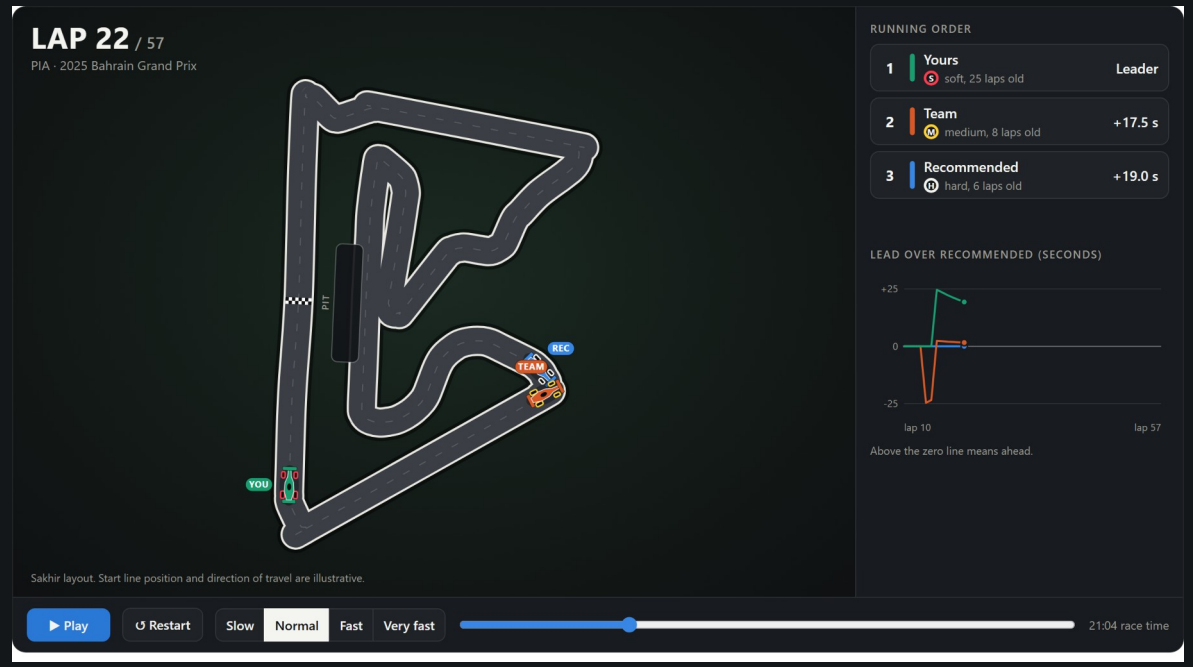
PREDICTIVE MODELLING · COURSE PROJECT

Can a model call the pit stop?

A race strategy simulator built on real Formula 1 lap data, 2022–2025



THE ANSWER FIRST

Race time is predictable; the exact pit lap is not

0.57%

Race-time error for the strategy really used: 24 s over a race of about 70 minutes

2.2×

Lower race-time error than constant pace: 24 s instead of 55 s

65%

Recommended number of stops matches the team's

8 laps

Average distance between the recommended first pit lap and the team's

Good strategies differ by about 2 seconds. That is smaller than the model's own error, so it can rank races, not pit laps.

DATA

Real race laps, split by season

Split	Seasons	Dry races	Laps used
Train	2022, 2023	35	24,039
Validation	2024	19	15,332
Test	2025	21	15,844

Source: FastF1. 92 races downloaded; 12 wet races dropped; 5 had no usable laps after cleaning.

Only clean laps are modelled: green flag, dry tyres, no pit entry or exit, within 4% of the driver's median pace.

Why split by season

Laps from one race share the same track, weather and car. A random split puts the same race on both sides and overstates accuracy.

Splitting by season means the model is always tested on races it has never seen.

2022–2025 is one set of technical rules, so the cars are comparable.

The 2025 season was used once, after the model was chosen.

Predict the change in pace, not the lap time

TARGET

delta % =
100 × (lap time ÷ baseline − 1)

Baseline = median of the driver's first clean laps (up to 8, between laps 2 and 12).

It is known a few laps into the race, so predicting later laps from it uses no future information.

Example: at a 90 s baseline, a lap predicted at +1.2% is 91.08 s.

Tyre age

Laps completed on the current set

Compound

Soft, medium or hard

Fuel burned

Share of race distance since the baseline

Race progress

Share completed, and race length

Baseline conditions

Tyre and track temperature when pace was set

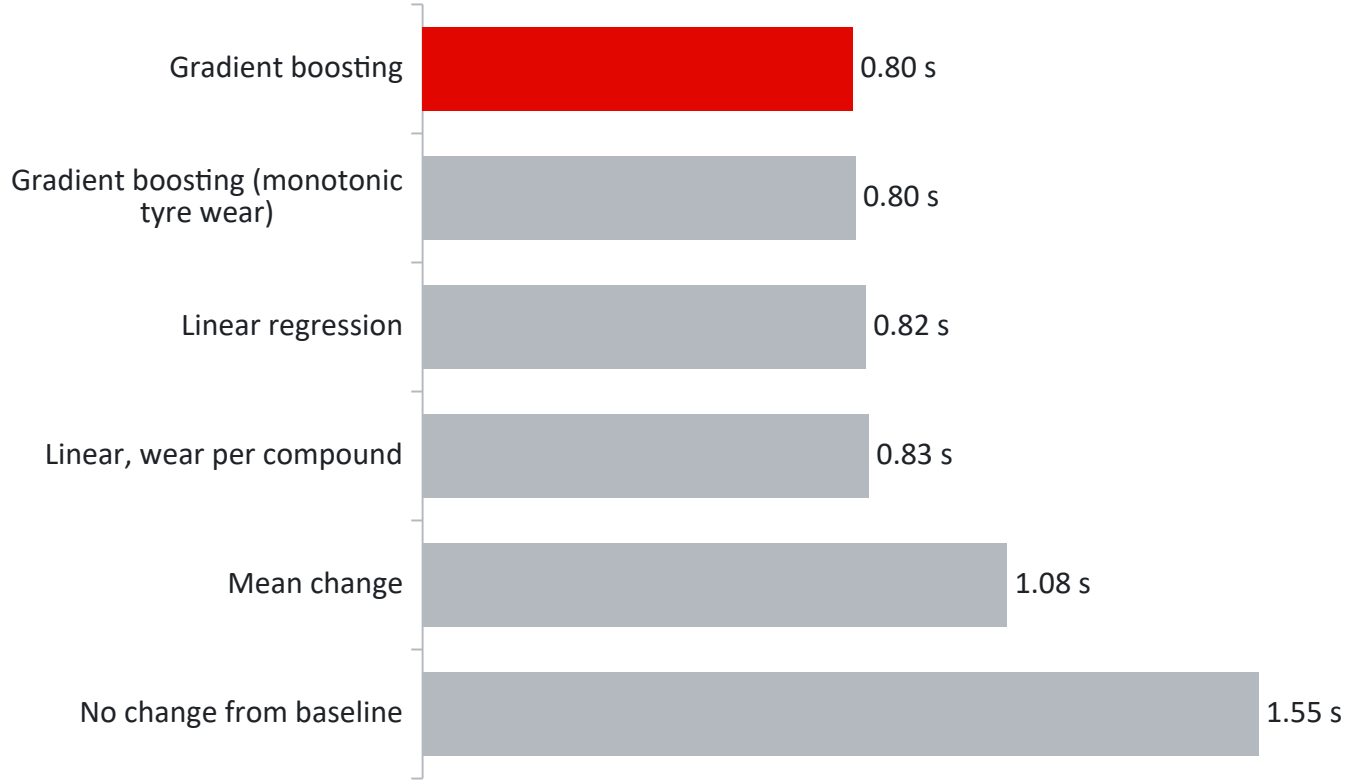
Circuit

Venue, as a category

MODEL COMPARISON

Small boosted trees narrowly beat linear regression

Validation RMSE, seconds per lap (2024 season)



What the comparison shows

Gradient boosting: 0.80 s per lap, selected.

Linear regression: 0.82 s. The gap is about 3%.

Assuming pace never changes: 1.55 s.

Trees with 31 leaves fitted race-specific noise and lost to linear regression. Trees with 4 leaves won.

TEST SEASON

On the unseen 2025 season the lap error is nearly halved

0.83 s

Lap error (RMSE) with the model

1.56 s

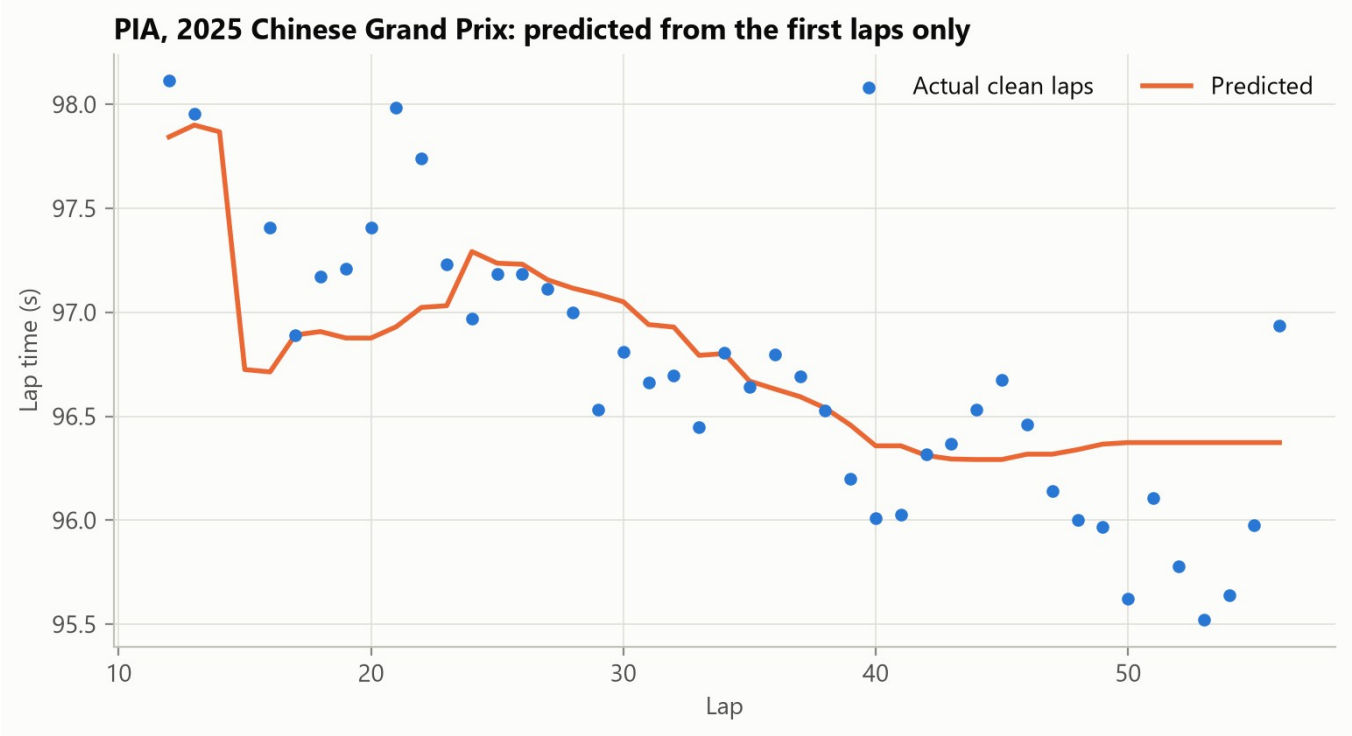
Lap error assuming constant pace

Where the error sits ($R^2 = 0.44$)

Driver over a whole race: 0.55% of pace. Shifts every strategy equally.

Stint within a race: 0.44%. This is what separates strategies.

Single lap: 0.65%. Mostly averages out.



One driver's race, predicted from their first clean laps only.

SIMULATOR

Turning the model into a race simulator

1

Tyre plan

Compound and tyre age for every lap of a strategy

2

Predict laps

Model output scaled by the driver's baseline pace

3

Add pit loss

Median green-flag loss at that circuit in past seasons

4

Search

Every one-stop plan and a grid of two-stop plans

5

Uncertainty

Random races drawn from stint and lap error

Rules the search enforces

At least two dry compounds in the race

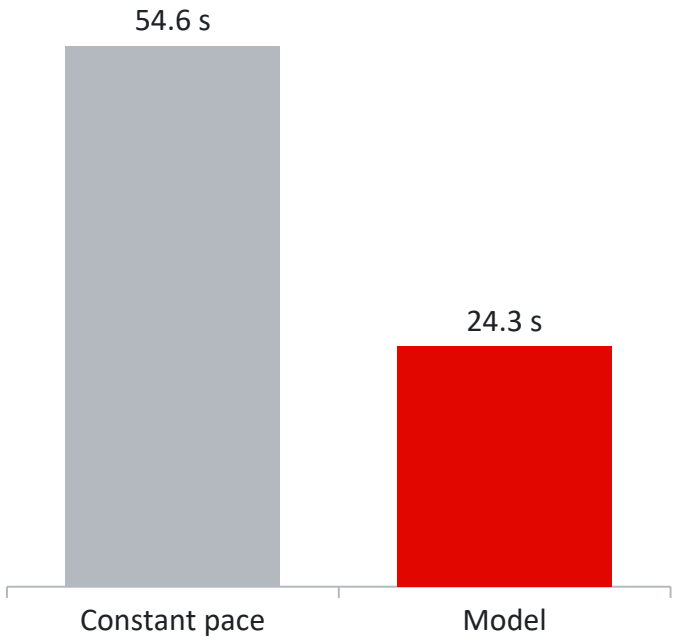
Every stint at least 5 laps long

No stint longer than tyres were seen to last at that circuit

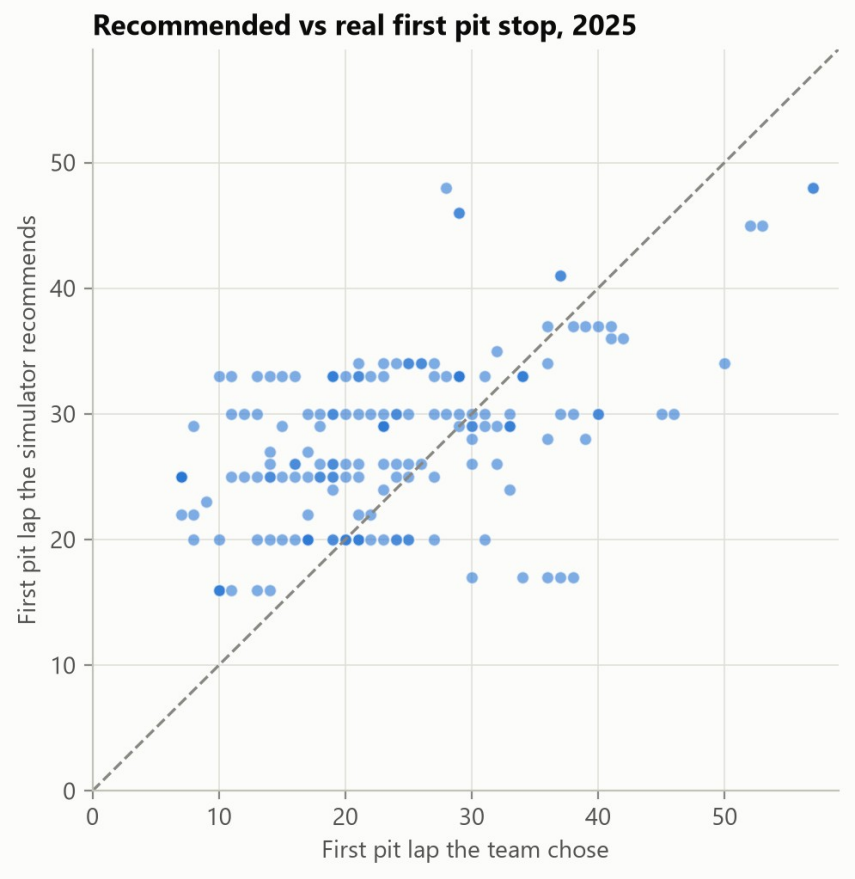
BACKTEST

Checked against 182 real driver-races from 2025

Race-time error (MAE), seconds



60 driver-races that stayed under green flags.



65%

same number of stops as the team

8.0 laps

average distance of the first pit lap

26%

of first pit laps within 3 laps

DEMO

The simulator as an animated race



Circuit outlines are real. Start-line position and direction of travel are illustrative.

1. **Three strategies race**

   Recommended, the team's real one, and your own

2. **Tyres and stops are visible**

   Wheel colour is the compound; cars use the pit box

3. **Live running order**

   Gaps, tyre age and a lead chart, lap by lap

4. **Random races**

   Each run draws from the model's measured error

LIMITATIONS

Why the pit lap is hard, and what would fix it

1

Survivorship in tyre data

Long stints are only seen when the tyres lasted, so old tyres look better than they are and stops come out too late.

2

No traffic or track position

Teams stop early to undercut a rival or rejoin in clear air. The simulator only finds the fastest plan in clean air.

3

No safety cars or weather

These decide many real races. Race time could only be checked on races that stayed green.

4

Gains are inside the error

Good strategies differ by about 2 s; stint-level error alone is larger. The team's choice is a reference, not proven best.

Next: add safety-car scenarios, separate the three compounds better, and model traffic.

CONCLUSION

An accurate model is not yet a decision

Lap time: 0.83 s error on an unseen season, about half the error of constant pace.

Race time: within 0.57% for the strategy really used.

Pit lap: the choice the simulator exists for turns on differences of about 2 seconds, which the model cannot resolve.

Validate a model at the level of the decision it will be used for.